

Split Conformal Prediction: Exchangeability, Ranks, and Finite-Sample Coverage

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Abstract

Conformal prediction provides finite-sample predictive guarantees without imposing a parametric model for the conditional distribution of the response. In the split conformal construction, the data are divided into a training sample, used to fit an arbitrary prediction rule, and a calibration sample, used to quantify predictive uncertainty. The probabilistic mechanism behind the method is elementary but powerful: exchangeability implies a symmetry of ranks, and that symmetry can be converted into a finite-sample coverage statement by a suitable order statistic of the calibration scores. This article develops that argument in detail. We first isolate the basic probabilistic facts concerning finite exchangeability, ranks, and order statistics. We then apply them to split conformal prediction, first for a general nonconformity score and then for the familiar absolute-residual score in regression. Particular attention is given to the finite-sample correction in the conformal quantile and to the distinction between marginal and conditional coverage. The presentation is deliberately organized around explicit definitions, lemmas, and proofs, so that the mathematical core can subsequently be translated into a formal proof environment such as Lean.

1 Introduction

A supervised learning problem starts from observations

$$Z_i = (X_i, Y_i) \in \mathcal{X} \times \mathcal{Y},$$

where X_i is a covariate and Y_i is the associated response. In classical statistical inference one often begins with a parametric family, estimates an unknown parameter, and then derives uncertainty statements from the sampling distribution of the estimator. Conformal prediction follows a different route. Its main object is not an unknown parameter but the response associated with a future covariate value. The goal is to construct, from observed data, a set-valued prediction rule

$$x \mapsto C(x) \subseteq \mathcal{Y}$$

such that a future response Y_{n+1} belongs to $C(X_{n+1})$ with a prescribed probability.

The basic target is therefore

$$\mathbb{P}\{Y_{n+1} \in C(X_{n+1})\} \geq 1 - \alpha, \tag{1}$$

where $\alpha \in (0, 1)$ is a chosen miscoverage level. What makes conformal prediction remarkable is that (1) can be obtained in finite samples, without Gaussian errors, linearity, homoskedasticity, or even a correctly specified prediction model. The essential structural assumption is instead *exchangeability*: the joint distribution of the observations is invariant under permutations.

In split conformal prediction, the observed sample is partitioned into two pieces. The first is used to fit a predictor $\hat{f}$, while the second is used only for calibration. Once the predictor has been fitted, each calibration observation is converted into a real-valued *nonconformity score*; large scores represent observations that fit the predictor poorly. The new observation is assigned

a score by the same rule. Conditional on the training sample, the calibration scores and the future score inherit exchangeability from the data. The problem is then reduced to a purely probabilistic statement about the rank of one exchangeable real-valued random variable among the others.

This reduction is the central theme of the article. Section 2 develops the probabilistic core: finite exchangeability, the uniform-rank principle, and its translation into an order-statistic probability. Section 3 applies the result to split conformal prediction, derives the standard finite-sample coverage theorem, specializes it to regression with absolute residuals, and closes with the distinction between marginal and pointwise conditional coverage.

Throughout, we emphasize the simplest setting in which the argument is completely transparent. In particular, the main rank calculation is first stated under a no-ties assumption. This is not essential for conformal prediction itself, but it removes a technical distraction and makes the logical structure especially clear. A short remark explains how ties modify the exact rank statement while preserving the conservative coverage inequality.

2 Probabilistic foundation: exchangeability, ranks, and order statistics

2.1 Finite exchangeability

We begin with the symmetry property that replaces independence in the conformal argument.

Definition 2.1 (Exchangeability). Let $W_1, \dots, W_N$ be random variables with values in a measurable space $\mathcal{W}$. The finite sequence $(W_1, \dots, W_N)$ is *exchangeable* if for every permutation π of $\{1, \dots, N\}$,

$$(W_1, \dots, W_N) \stackrel{d}{=} (W_{\pi(1)}, \dots, W_{\pi(N)}).$$

Equivalently, for every measurable set $A \subseteq \mathcal{W}^N$,

$$\mathbb{P}((W_1, \dots, W_N) \in A) = \mathbb{P}((W_{\pi(1)}, \dots, W_{\pi(N)}) \in A).$$

Every i.i.d. sample is exchangeable, but exchangeability is weaker than independence. For conformal prediction this distinction matters: the validity argument uses permutation symmetry, not a factorization of the joint distribution into independent marginals.

A basic closure property will be useful later.

Lemma 2.2 (Coordinatewise transformations preserve exchangeability). *Let $W_1, \dots, W_N$ be exchangeable, and let $g : \mathcal{W} \rightarrow \mathcal{S}$ be measurable. Then*

$$g(W_1), \dots, g(W_N)$$

is exchangeable.

Proof. For any permutation π ,

$$(W_1, \dots, W_N) \stackrel{d}{=} (W_{\pi(1)}, \dots, W_{\pi(N)}).$$

Applying the measurable map

$$(w_1, \dots, w_N) \mapsto (g(w_1), \dots, g(w_N))$$

to both sides preserves equality in distribution, giving

$$(g(W_1), \dots, g(W_N)) \stackrel{d}{=} (g(W_{\pi(1)}), \dots, g(W_{\pi(N)})).$$

□

The same idea applies conditionally. If a random object T is fixed and, conditional on T , the variables $W_1, \dots, W_N$ are exchangeable, then any common transformation g_T applied to all coordinates produces conditionally exchangeable transformed variables. This conditional form is exactly what is needed in split conformal prediction, where T will be the training sample and g_T will be the score map determined by the fitted predictor.

2.2 Ranks under exchangeability

Let $R_1, \dots, R_N$ be real-valued random variables. Assume for the moment that ties occur with probability zero:

$$\mathbb{P}(R_i = R_j) = 0, \quad i \neq j. \quad (2)$$

Then exactly one variable occupies each position in the ordered sample.

Definition 2.3 (Rank). Under (2), the rank of R_j among $R_1, \dots, R_N$ is

$$\text{rank}(R_j) := 1 + \sum_{i \neq j} \mathbf{1}\{R_i < R_j\}. \quad (3)$$

Thus $\text{rank}(R_j) = 1$ if R_j is the smallest observation, and $\text{rank}(R_j) = N$ if it is the largest.

The key probabilistic fact is that exchangeability makes every coordinate equally likely to occupy every rank.

Theorem 2.4 (Uniform-rank principle). *Let $R_1, \dots, R_N$ be exchangeable real-valued random variables satisfying the no-ties condition (2). Then, for every fixed index $j \in \{1, \dots, N\}$,*

$$\text{rank}(R_j) \sim \text{Unif}\{1, \dots, N\}.$$

Equivalently,

$$\mathbb{P}(\text{rank}(R_j) = k) = \frac{1}{N}, \quad k = 1, \dots, N. \quad (4)$$

Proof. Fix $k \in \{1, \dots, N\}$. Because ties do not occur, exactly one of the N variables has rank k . Hence, almost surely,

$$\sum_{j=1}^N \mathbf{1}\{\text{rank}(R_j) = k\} = 1.$$

Taking expectations gives

$$\sum_{j=1}^N \mathbb{P}(\text{rank}(R_j) = k) = 1. \quad (5)$$

By exchangeability, the probability in (5) is the same for every j : no coordinate is probabilistically distinguished from any other. If that common value is denoted by p_k , then $Np_k = 1$, so $p_k = 1/N$. Since this holds for every k , the rank is uniform on $\{1, \dots, N\}$. $\square$

Remark 2.5 (Why independence is not used). The proof never factors a joint probability into a product of marginal probabilities. It uses only two ingredients: permutation invariance and the deterministic fact that, in the absence of ties, one and only one observation occupies a given rank. This is why exchangeability, rather than independence, is the natural assumption for the conformal rank argument.

For later use, we specialize to $N = m + 1$ and single out the last observation.

Corollary 2.6. *Let $R_1, \dots, R_m, R_{m+1}$ be exchangeable and almost surely distinct. Then for every $k \in \{1, \dots, m + 1\}$,*

$$\mathbb{P}(\text{rank}(R_{m+1}) \leq k) = \frac{k}{m + 1}. \quad (6)$$

Proof. By Theorem 2.4, each rank $1, \dots, m + 1$ has probability $1/(m + 1)$. Summing the first k probabilities yields (6). $\square$

2.3 From ranks to order statistics

Let

$$R_{(1)} \leq R_{(2)} \leq \cdots \leq R_{(m)}$$

denote the order statistics of the first m variables, which we will later interpret as calibration scores. The next proposition makes explicit the bridge between the rank of the $(m+1)$ -st score and a threshold based only on the first m scores.

Proposition 2.7 (Rank–order-statistic equivalence). *Under the assumptions of Corollary 2.6, for every $k \in \{1, \dots, m\}$,*

$$R_{m+1} \leq R_{(k)} \iff \text{rank}(R_{m+1}) \leq k. \quad (7)$$

Consequently,

$$\mathbb{P}(R_{m+1} \leq R_{(k)}) = \frac{k}{m+1}. \quad (8)$$

Proof. Because there are no ties, $\text{rank}(R_{m+1}) \leq k$ means that at most $k-1$ of the calibration scores $R_1, \dots, R_m$ are strictly smaller than R_{m+1} . This is equivalent to saying that R_{m+1} does not exceed the k -th smallest calibration score, namely $R_{(k)}$. The probability formula then follows from Corollary 2.6. $\square$

Now fix a target miscoverage level $\alpha \in (0, 1)$ and define

$$k_\alpha := \lceil (m+1)(1-\alpha) \rceil. \quad (9)$$

If $k_\alpha \leq m$, then $R_{(k_\alpha)}$ is an ordinary order statistic of the calibration sample and

$$\mathbb{P}(R_{m+1} \leq R_{(k_\alpha)}) = \frac{k_\alpha}{m+1} \geq 1-\alpha. \quad (10)$$

The appearance of $m+1$, rather than m , is the characteristic finite-sample correction in split conformal prediction.

Definition 2.8 (Conformal calibration quantile). Assume $k_\alpha \leq m$. The split conformal calibration threshold at level $1-\alpha$ is

$$\hat{q}_{1-\alpha} := R_{(k_\alpha)}, \quad k_\alpha = \lceil (m+1)(1-\alpha) \rceil. \quad (11)$$

It is useful to compare (11) with the ordinary empirical $(1-\alpha)$ -quantile, which is often based on an index near $m(1-\alpha)$. Conformal prediction deliberately moves the index upward from the m -sample scale to the $(m+1)$ -sample scale, because the future score must be inserted symmetrically among the m calibration scores. The correction is small for large m but essential for an exact finite-sample argument.

Remark 2.9 (The extreme small- α case). If

$$\alpha < \frac{1}{m+1},$$

then $k_\alpha = m+1$, which is not an order-statistic index available from only m calibration scores. One standard convention is to set $\hat{q}_{1-\alpha} = +\infty$, yielding the trivial full prediction set and therefore coverage one. For a first formalization, it is often cleaner to assume

$$\alpha \geq \frac{1}{m+1},$$

so that $k_\alpha \leq m$ and all thresholds remain real-valued calibration order statistics.

Remark 2.10 (Ties). The exact equality (8) uses the no-ties assumption. If ties are possible, the rank is no longer uniquely described by (3). One may introduce randomized tie-breaking to recover an exactly uniform rank. Alternatively, using the weak inequality in the usual conformal construction leads to a conservative statement of the form

$$\mathbb{P}(R_{m+1} \leq \hat{q}_{1-\alpha}) \geq 1 - \alpha.$$

Thus the no-ties assumption simplifies the exposition but is not fundamental to conformal validity.

3 Split conformal prediction

3.1 Training, calibration, and the prediction problem

We now return to supervised learning. Let

$$Z_i = (X_i, Y_i) \in \mathcal{X} \times \mathcal{Y}, \quad i = 1, \dots, n+1,$$

and suppose that the complete sequence

$$Z_1, \dots, Z_n, Z_{n+1}$$

is exchangeable. The observation $Z_{n+1} = (X_{n+1}, Y_{n+1})$ is the future point for which only X_{n+1} will be observed at prediction time.

Choose integers $n_{\text{tr}} \geq 1$ and $m \geq 1$ with

$$n = n_{\text{tr}} + m.$$

The observed sample is split into

$$\underbrace{Z_1, \dots, Z_{n_{\text{tr}}}}_{\text{training sample}} \quad \text{and} \quad \underbrace{Z_{n_{\text{tr}}+1}, \dots, Z_n}_{\text{calibration sample}}.$$

To lighten notation in the calibration argument, we may relabel the calibration observations as

$$\tilde{Z}_1, \dots, \tilde{Z}_m,$$

and the future observation as $\tilde{Z}_{m+1}$. Nothing probabilistic is changed by this relabeling.

Let $\mathcal{A}$ be an arbitrary learning algorithm. From the training sample it produces a prediction rule

$$\hat{f} = \mathcal{A}(Z_1, \dots, Z_{n_{\text{tr}}}).$$

For regression, $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}$; for a more general response space, $\hat{f}$ may represent any fitted predictive object needed to define a score.

The split conformal method does not require $\hat{f}$ to be unbiased, consistent, correctly specified, or even especially accurate. Accuracy affects the size of the final prediction set, but the coverage argument depends on symmetry of the scores rather than on the statistical quality of the fitted model.

3.2 Nonconformity scores

A *nonconformity score* measures how poorly a candidate pair (x, y) agrees with the fitted predictor. Formally, once $\hat{f}$ is fixed, let

$$S(\cdot, \cdot; \hat{f}) : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$$

be measurable, with larger values interpreted as greater nonconformity.

For each calibration observation define

$$R_i := S(\tilde{X}_i, \tilde{Y}_i; \hat{f}), \quad i = 1, \dots, m, \quad (12)$$

and for the future observation define

$$R_{m+1} := S(\tilde{X}_{m+1}, \tilde{Y}_{m+1}; \hat{f}). \quad (13)$$

The following lemma is the bridge from the data to the probabilistic theory of Section 2.

Lemma 3.1 (Conditional exchangeability of split-conformal scores). *Assume that*

$$Z_1, \dots, Z_n, Z_{n+1}$$

is exchangeable and that $\hat{f}$ is a measurable function of the training sample only. Then, conditional on the training sample, the calibration observations together with the future observation are exchangeable. Consequently, conditional on the training sample, the scores

$$R_1, \dots, R_m, R_{m+1}$$

are exchangeable.

Proof. Fix a permutation π of the indices corresponding to the m calibration observations and the future observation, while leaving the training indices unchanged. Since the full sequence is exchangeable, the joint distribution is invariant under this permutation. Therefore, after conditioning on the training coordinates, the remaining $m + 1$ coordinates are still symmetric under permutations among themselves. This yields conditional exchangeability of

$$\tilde{Z}_1, \dots, \tilde{Z}_m, \tilde{Z}_{m+1}.$$

Given the training sample, the fitted predictor $\hat{f}$ is fixed. The same measurable map

$$(x, y) \mapsto S(x, y; \hat{f})$$

is applied to every remaining observation. Lemma 2.2, applied conditionally, therefore gives conditional exchangeability of the scores. $\square$

Remark 3.2 (Why the split is essential to this simple proof). The score function is fitted using only the training sample, so after conditioning on that sample the same fixed scoring rule is applied to every calibration point and to the future point. If the calibration responses were also used to fit $\hat{f}$, this simple coordinatewise symmetry argument would no longer apply directly. Full conformal prediction restores symmetry by refitting or otherwise treating the candidate future response symmetrically, but split conformal prediction avoids that computational cost by separating fitting from calibration.

3.3 The conformal quantile and prediction set

Let

$$R_{(1)} \leq \dots \leq R_{(m)}$$

be the order statistics of the calibration scores in (12). Fix $\alpha \in (0, 1)$ and define

$$k_\alpha := \lceil (m + 1)(1 - \alpha) \rceil. \quad (14)$$

For the main theorem we assume $k_\alpha \leq m$, equivalently $\alpha \geq 1/(m + 1)$, and set

$$\hat{q}_{1-\alpha} := R_{(k_\alpha)}. \quad (15)$$

For a new covariate value $x \in \mathcal{X}$, define the split conformal prediction set

$$C(x) := \{y \in \mathcal{Y} : S(x, y; \hat{f}) \leq \hat{q}_{1-\alpha}\}. \quad (16)$$

The construction is conceptually simple: a candidate response y is accepted if its nonconformity score is no larger than the calibrated threshold.

By definition,

$$\tilde{Y}_{m+1} \in C(\tilde{X}_{m+1}) \iff S(\tilde{X}_{m+1}, \tilde{Y}_{m+1}; \hat{f}) \leq \hat{q}_{1-\alpha} \quad (17)$$

$$\iff R_{m+1} \leq R_{(k_\alpha)}. \quad (18)$$

Thus predictive coverage is exactly the order-statistic event analyzed in the previous section.

Theorem 3.3 (Finite-sample validity of split conformal prediction). *Suppose that*

$$Z_1, \dots, Z_n, Z_{n+1}$$

is exchangeable. Let $\hat{f}$ be fitted using only the training sample, let S be a measurable nonconformity score, and let C be defined by (16) with

$$k_\alpha = \lceil (m+1)(1-\alpha) \rceil \leq m.$$

Assume, for the exact rank calculation, that conditional on the training sample the $m+1$ scores are almost surely distinct. Then

$$\mathbb{P}\{Y_{n+1} \in C(X_{n+1})\} \geq 1 - \alpha. \quad (19)$$

More precisely, conditional on the training sample,

$$\mathbb{P}\{Y_{n+1} \in C(X_{n+1}) \mid \mathcal{T}\} = \frac{k_\alpha}{m+1} \geq 1 - \alpha \quad (20)$$

for almost every training sample $\mathcal{T}$.

Proof. Condition on the training sample $\mathcal{T}$. By Lemma 3.1, the scores

$$R_1, \dots, R_m, R_{m+1}$$

are exchangeable conditional on $\mathcal{T}$. Under the no-ties assumption, Proposition 2.7 therefore gives

$$\mathbb{P}(R_{m+1} \leq R_{(k_\alpha)} \mid \mathcal{T}) = \frac{k_\alpha}{m+1}.$$

By construction of k_α ,

$$k_\alpha = \lceil (m+1)(1-\alpha) \rceil \geq (m+1)(1-\alpha),$$

so

$$\frac{k_\alpha}{m+1} \geq 1 - \alpha.$$

Finally,

$$\{Y_{n+1} \in C(X_{n+1})\} = \{R_{m+1} \leq R_{(k_\alpha)}\},$$

which proves (20). Taking expectations over the training sample yields the unconditional, or marginal, guarantee (19). $\square$

Remark 3.4 (Distribution-free does not mean assumption-free). The theorem is often described as distribution-free because it does not assume a parametric form for the distribution of (X, Y) . Nevertheless, it is not assumption-free. Exchangeability is the structural hypothesis that makes the rank argument possible. If exchangeability fails, for example because of time dependence or distribution shift, the finite-sample guarantee need not remain valid without additional modifications.

3.4 Regression with absolute residuals

The most familiar split conformal interval arises in regression, where $\mathcal{Y} = \mathbb{R}$ and the score is the absolute residual

$$S(x, y; \hat{f}) := |y - \hat{f}(x)|. \quad (21)$$

Then the calibration scores are

$$R_i = |\tilde{Y}_i - \hat{f}(\tilde{X}_i)|, \quad i = 1, \dots, m. \quad (22)$$

Let

$$\hat{q}_{1-\alpha} = R_{(k_\alpha)}.$$

For a candidate response y at covariate value x ,

$$\begin{aligned} y \in C(x) &\iff |y - \hat{f}(x)| \leq \hat{q}_{1-\alpha} \\ &\iff \hat{f}(x) - \hat{q}_{1-\alpha} \leq y \leq \hat{f}(x) + \hat{q}_{1-\alpha}. \end{aligned}$$

Hence

$$C(x) = [\hat{f}(x) - \hat{q}_{1-\alpha}, \hat{f}(x) + \hat{q}_{1-\alpha}]. \quad (23)$$

Corollary 3.5 (Split conformal regression interval). *Under the assumptions of Theorem 3.3, with the absolute-residual score (21), the interval (23) satisfies*

$$\mathbb{P}\{Y_{n+1} \in [\hat{f}(X_{n+1}) - \hat{q}_{1-\alpha}, \hat{f}(X_{n+1}) + \hat{q}_{1-\alpha}]\} \geq 1 - \alpha.$$

The interval has constant half-width $\hat{q}_{1-\alpha}$ as a function of x . This is a consequence of the choice of score, not of conformal prediction in general. More adaptive scores can produce prediction intervals whose width varies with the covariate. The validity proof, however, continues to be driven by the same exchangeability-and-rank mechanism as long as the score is applied symmetrically to the calibration and future observations after the training stage has been fixed.

3.5 Marginal versus conditional coverage

The guarantee

$$\mathbb{P}\{Y_{n+1} \in C(X_{n+1})\} \geq 1 - \alpha \quad (24)$$

is called a *marginal coverage* statement. The probability averages over the randomness of the new covariate X_{n+1} , the new response Y_{n+1} , and the data used to construct the prediction set. As Theorem 3.3 shows, in the standard split construction the argument can actually be made conditional on the training sample. But this should not be confused with conditioning on the new covariate value.

A much stronger requirement would be the pointwise conditional statement

$$\mathbb{P}\{Y_{n+1} \in C(X_{n+1}) \mid X_{n+1} = x\} \geq 1 - \alpha \quad \text{for every } x \in \mathcal{X}. \quad (25)$$

The marginal guarantee (24) does not imply (25). A procedure can have correct overall coverage while undercovering in some regions of the covariate space and overcovering in others.

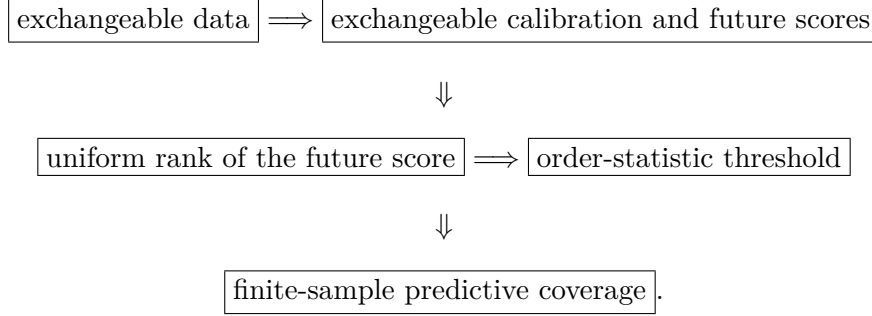
This distinction is fundamental. Standard conformal prediction converts the global permutation symmetry encoded by exchangeability into a global predictive coverage statement. Exact, distribution-free, pointwise conditional coverage is generally unattainable without additional assumptions or without allowing prediction sets to become uninformative. Therefore one should keep separate three questions:

- (i) *validity*: does the method attain the required marginal coverage?;
- (ii) *efficiency*: how small or informative are the resulting prediction sets?;
- (iii) *conditional behavior*: how evenly is coverage distributed over the covariate space?

Only the first question is settled automatically by the basic split conformal theorem under exchangeability.

3.6 Logical structure of the argument

The entire theory developed above can be summarized as the following chain:



The statistical learning algorithm enters only through the score map. Once the training sample has been fixed, the coverage proof is a finite probabilistic statement about symmetry and ordering. This modularity is precisely what makes split conformal prediction attractive both conceptually and for formal verification: the learning procedure can remain a black box, while the validity theorem reduces to a small collection of reusable probabilistic lemmas.

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